

Yuru Feng

619-953-7267 | Wechat: YuruF25 | Email: yuf043@ucsd.edu | scholar.google.com/yurufeng

EDUCATION

University of California, San Diego

Master of Data Science

San Diego, CA

Sep. 2025 - Jun. 2027(expected)

Southern University of Science and Technology

Bachelor of Science in Data Science and Big Data Technology

Shenzhen, China

Sep. 2021 - Jun. 2025

- GPA: 3.8/4.0

PUBLICATIONS

Yaoqi Chen, Haibin Lai, **Yuru Feng**, Chuyu Han, Qianxi Zhang, Baotong Lu, Menghao Li, Xinjiang Wang, Zewen Jin, Hao Wu, Cheng Li, Qi Chen, "**Beyond Semantic Organization: Memory as Execution State Management for Long-Horizon Agents**," *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026. | arXiv:2606.06090

Yuru Feng, Yaoqi Chen, Beidi Zhao, Qianxi Zhang, Xinjiang Wang, Jianan Lu, Zhirui Wang, Shusen Xu, Zengzhong Li, Qi Chen, "**SkillHEX: Improving Agent skills via Hypothesis-Driven Autonomous Exploration and Exploitation**" (AAAI2027 Under review) | arXiv:2608.05628

Beidi Zhao, Yaoqi Chen, **Yuru Feng**, Menghao Li, Qianxi Zhang, Baotong Lu, Jianan Lu, Zhirui Wang, Xinjiang Wang, Shusen Xu, Zengzhong Li, Xiaoxiao Li, Qi Chen, "**MESA: Task-Adaptive Multi-Structure Evidence Selection for Long-Horizon Agent Memory**" (AAAI2027 Under review) | arXiv:2608.10108

WORK EXPERIENCE

Research Intern | Microsoft Research Asia(MSRA)

Mar. 2026 – Present

System Research Group(AI memory)

Beijing, China

- Led research on self-evolving agent skills, developing a hypothesis-driven framework that diagnoses failure causes from sparse feedback and guides autonomous skill refinement through adaptive exploration and exploitation; improved task performance by 9.5 pp while reducing token consumption by 18%.
- Investigated failure modes of retrieval-based memory for long-horizon agents, identifying state fragmentation as a major source of execution inconsistency; co-developed a hierarchical execution-state memory with compression, verification, and rollback, improving task success by 7.8–20.4 pp while reducing token usage by 55.1% versus full-context methods.
- Studied task-adaptive memory access across heterogeneous memory structures and contributed to a structure-level selection mechanism that dynamically combines complementary evidence under an exploration–exploitation strategy; improved performance by 8.5 pp over the strongest baseline while reducing token consumption by 41%.

Artificial Intelligence Engineer Intern | MGI Tech Co., Ltd

Mar. 2025 – Aug. 2025

Frontier Technology R&D X Center

Shenzhen, China

- Developed a domain-specific LLM post-training pipeline for biomedical knowledge extraction using DeepSeek-R1 reasoning-data distillation, LoRA fine-tuning, and GRPO, and built a LangGraph-based multi-agent workflow with human-in-the-loop adaptive prompt optimization.
- Designed the biomedical agent's RAG and knowledge-graph pipeline, integrating VLM/OCR document parsing, semantics-aware chunking, structured retrieval, evidence tracing, and multi-hop reasoning; achieved a 7.5% improvement in core product metrics while substantially reducing the cost of standard GraphRAG-based solutions.

SELECTED HONORS & AWARDS

Excellent Graduates of Southern University of Science and Technology(2025)

Outstanding Student Scholarship, Zhiren College, SUSTech(2024)

Excellence in Student Leadership Award, SUSTech(2023)